REPORT: Building a Network Attack Monitoring and Detection System using ELK Stack

1. Introduction

In this project, a basic Security Information and Event Management (SIEM) system was implemented using the ELK Stack, which includes:

- Elasticsearch
- Logstash
- Kibana

The system is designed to collect, process, analyze, and visualize log data from servers in order to detect potential security threats such as brute-force attacks and unauthorized access attempts.

Attacker (Kali) → Victim (Ubuntu) → Filebeat → Logstash → Elasticsearch → Kibana

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2. Objectives

The main objectives of this project are:

- Collect system logs from a Linux server.
- Analyze logs to detect suspicious activities.
- Simulate network attacks such as SSH brute-force.
- Visualize logs and security events using Kibana dashboards.

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3. System Architecture

The system consists of the following components:

- **Target Machine (Ubuntu Server)**
  Used to generate system logs (e.g., SSH login attempts).

- **Attacker Machine (Kali Linux)**
  Used to simulate attacks (e.g., brute-force SSH using Hydra).
- **SIEM Server (Ubuntu + ELK Stack)**
  Responsible for log collection, processing, storage, and visualization.

Workflow:

1. Logs are generated on the target machine.
2. Filebeat collects logs from /var/log/auth.log.
3. Logs are sent to Logstash via port 5044.
4. Logstash processes and forwards logs to Elasticsearch.
5. Kibana displays logs in real-time.

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4. Implementation Steps

4.1 Install ELK Stack

- Installed Elasticsearch, Logstash, and Kibana on Ubuntu.
- Configured services and verified connectivity.
- Note that when you load Elasticsearch -y, the security autoconfiguration will prompt you for a password.
- My password here is kLaJ53tOPtjxPj2M2se0

```
tien@tien:~$ curl -k -u elastic:kLaJ53tOPtjxPj2M2se0 https://localhost:9200
{
  "name" : "tien",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "paZHU2SpSXKGBble7_e1gA",
  "version" : {
    "number" : "8.19.13",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "f6279a2167a248af2bb287409231fb0cf52aa0db",
    "build_date" : "2026-03-16T10:08:32.007626438Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.2",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

```
tien@tien:~$ sudo systemctl enable elasticsearch
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /lib/systemd/system/elasticsearch.service.
tien@tien:~$ sudo systemctl start elasticsearch
tien@tien:~$ sudo systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
     Loaded: loaded (/lib/systemd/system/elasticsearch.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2026-04-07 15:11:30 +07; 39s ago
       Docs: https://www.elastic.co
   Main PID: 12219 (java)
      Tasks: 100 (limit: 4549)
     Memory: 2.3G
        CPU: 32.321s
     CGroup: /system.slice/elasticsearch.service
             ├─12219 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name>
             ├─12278 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.netw>
             └─12298 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller
```

```
tien@tien:~$ sudo systemctl enable kibana
Created symlink /etc/systemd/system/multi-user.target.wants/kibana.service → /lib/systemd/system/kibana.service.
tien@tien:~$ sudo systemctl start kibana
tien@tien:~$ sudo /usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana
eyJ2ZXIiOiI4LjE0LjAiLCJhZHIiOlsiMTkyLjE2OC4xMy4xNDM6OTIwMCJdLCJmZ3IiOiJkMWFlOTBhZjZjMTIwMGIzYjdmMjgzNGU4OTY4YzM0
iwia2V5IjoicjRrR1o1MEJpaVY0cjhMbEVwME46Q052X1g5MWh3SG94b3ZwLUFpT2FNZyJ9
tien@tien:~$ sudo systemctl status kibana
● kibana.service - Kibana
     Loaded: loaded (/lib/systemd/system/kibana.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2026-04-07 15:18:50 +07; 4min 13s ago
       Docs: https://www.elastic.co
   Main PID: 12839 (node)
      Tasks: 11 (limit: 4549)
     Memory: 302.8M
        CPU: 7.383s
     CGroup: /system.slice/kibana.service
             └─12839 /usr/share/kibana/bin/../node/glibc-217/bin/node /usr/share/kibana/bin/../src/cl>
```

4.2 Configure Log Collection

- Installed Filebeat on the target machine.
- Enabled system module:

sudo filebeat modules enable system

- Configured Filebeat to send logs to Logstash:

output.logstash:

hosts: ["192.168.13.143:5044"]

```
tien@tien:~$ sudo filebeat modules enable system
Module system is already enabled
tien@tien:~$ sudo nano /etc/filebeat/modules.d/system.yml
tien@tien:~$ sudo systemctl restart filebeat
tien@tien:~$ sudo systemctl status filebeat
● filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch.
     Loaded: loaded (/lib/systemd/system/filebeat.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2026-04-07 16:24:33 +07; 3s ago
       Docs: https://www.elastic.co/beats/filebeat
   Main PID: 6648 (filebeat)
      Tasks: 8 (limit: 4549)
     Memory: 53.7M
        CPU: 144ms
     CGroup: /system.slice/filebeat.service
             └─6648 /usr/share/filebeat/bin/filebeat --environment systemd -c /etc/filebeat/
```

```
# ---------------------------- Elasticsearch Output ----------------------------
#output.elasticsearch:
  # Array of hosts to connect to.
 # hosts: ["localhost:9200"]

  # Performance preset - one of "balanced", "throughput", "scale",
  # "latency", or "custom".
 # preset: balanced

  # Protocol - either `http` (default) or `https`.
  #protocol: "https"

  # Authentication credentials - either API key or username/password.
  #api_key: "id:api_key"
  #username: "elastic"
  #password: "changeme"

# ------------------------------ Logstash Output -------------------------------
output.logstash:
  # The Logstash hosts
  hosts: ["192.168.13.143:5044"]

  # Optional SSL. By default is off.
  # List of root certificates for HTTPS server verifications
  #ssl.certificate_authorities: ["/etc/pki/root/ca.pem"]

  # Certificate for SSL client authentication
  #ssl.certificate: "/etc/pki/client/cert.pem"

  # Client Certificate Key
  #ssl.key: "/etc/pki/client/cert.key"
```

4.3 Configure Logstash

Created pipeline:

sudo nano /etc/logstash/conf.d/logstash.conf

```
input {

  beats {

    port => 5044

  }

}


filter {
```

```
  # optional parsing
}

output {
  elasticsearch {
    hosts => ["https://localhost:9200"]
    user => "elastic"
    password => "your_password"
    ssl_certificate_verification => false
    index => "security-logs-%{+YYYY.MM.dd}"
  }
}
```

4.4 Simulate Attack

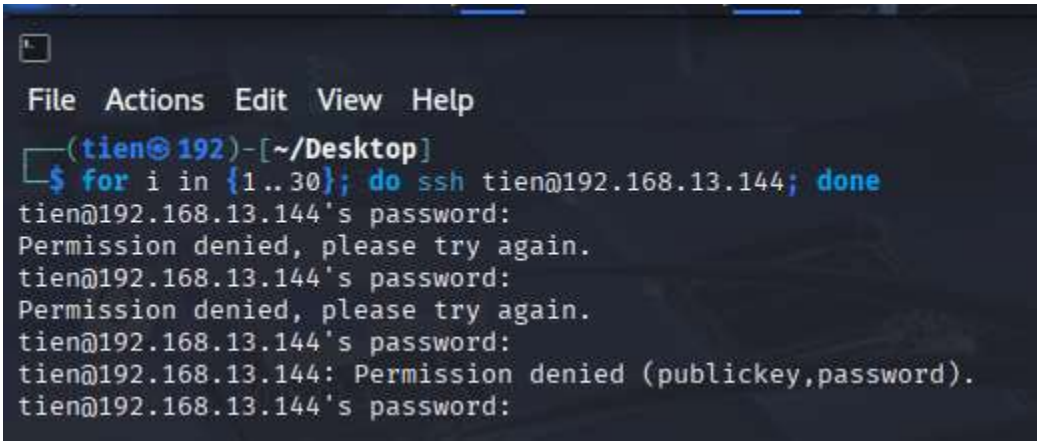
Used Kali Linux to perform brute-force attack:

for i in {1..30}; do ssh tien@192.168.13.144; done

or:

hydra -l tien -P rockyou.txt ssh://192.168.13.144 -t 4

This generated multiple failed login attempts.



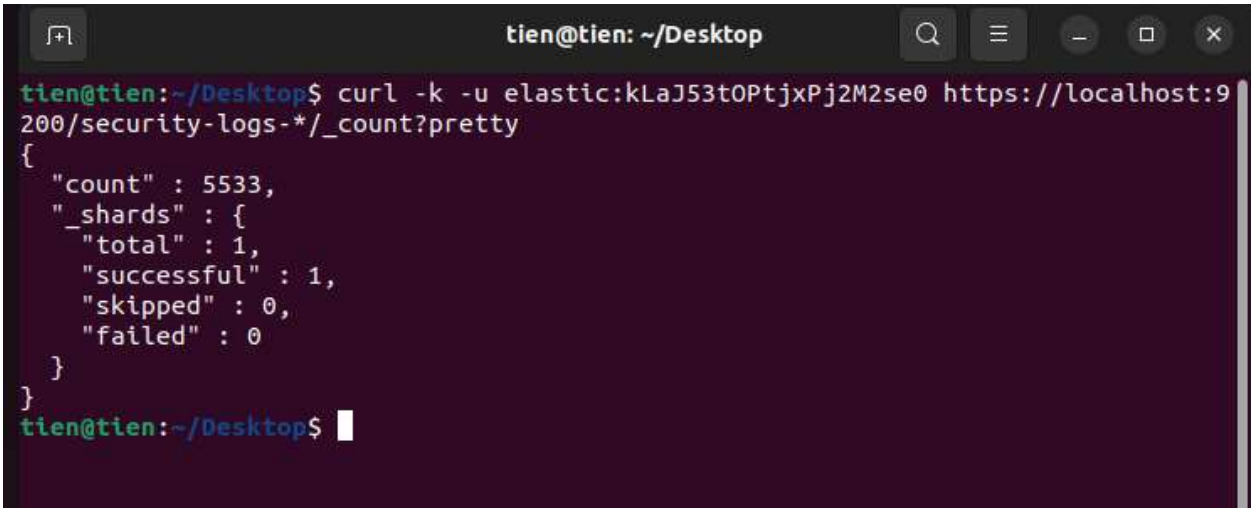
4.5 Verify Logs in Elasticsearch

curl -k -u elastic:<password> https://localhost:9200/security-logs-*/_count?pretty

Result:

"count" : 5533

Logs were successfully ingested.



4.6 Visualize Logs in Kibana

- Created Data View: security-logs-*
- Selected time field: @timestamp
- Viewed logs in Discover tab

Example query:

message: "Failed password"

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5. Results

The system successfully:

- Collected system logs from Ubuntu server
- Detected brute-force SSH attacks
- Stored logs in Elasticsearch
- Visualized logs in Kibana

Example detected logs:

- Failed SSH login attempts
- Source IP address of attacker
- Timestamp of attack

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6. Challenges and Solutions

Issue 1: Kibana Timeout

- Cause: Large dataset and limited VM resources
- Solution:
    - Increased RAM to 8GB

    - Reduced query time range
    - Increased request timeout

Issue 2: Unassigned Shards (RED status)

- Cause: Single-node cluster with replica shards
- Solution:
    - Removed replica or reallocated shard
    - Deleted corrupted index

Issue 3: Filebeat Service Failure

- Cause: Misconfiguration in output
- Solution:
    - Verified configuration
    - Restarted service

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7. Conclusion

This project demonstrates how a SIEM system can be built using the ELK Stack to monitor and detect security threats in a network.

The system effectively detects brute-force attacks and provides real-time visualization, making it a useful tool for security monitoring and incident response.